



UNIVERSITI TEKNIKAL MALAYSIA MELAKA
FACULTY OF INFORMATION AND COMMUNICATION
TECHNOLOGY

PROJEK SARJANA MUDA 1: PROPOSAL FORM

A TITLE OF PROPOSED PROJECT | TAJUK PROJEK YANG DICADANGKAN

INTELLIGENT RESUME SCREENING AND CANDIDATE RANKING SYSTEM FOR RECRUITMENT DECISION SUPPORT

B DETAILS OF STUDENT | BUTIRAN PELAJAR

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C PROJECT INFORMATION | MAKLUMAT PROJEK

(i) Executive Summary of Project Proposal [Maximum 300 words]

(Please include the background of the project, problem statements, objectives, and expected outcomes/ proposed solution from the project)

Recruitment is one of the critical activity in human resource management as it directly affects organizational performance and long-term workforce quality. While today's employers are often inundated with large number of applicants, may find it increasingly challenging to pinpoint suitable candidates. Human resource (HR) departments are responsible for multiple operational tasks, the manual gathering, sorting and screening of resumes becomes a time-consuming process that slow down recruitment decisions. Although most online job platforms provide automated candidate recommendations, these suggestions are often too general and may not accurately reflect the specific requirements stated in job descriptions. As a result, HR recruiters still need to manually review many resumes to determine candidate suitability. This project proposes the development of an Intelligent Resume Screening and Explainable Candidate Ranking System that assists HR recruiters during the early stage of recruitment. This system allows HR to upload job descriptions from which key job requirements such as skills, educational qualification, experience and job scope and responsibilities are extracted. The system then generates a unique application link where candidate can submit their applications and upload resumes directly into the system database. The system processes resumes using automated text extraction and evaluate candidates using a weighted similarities scoring model based on how well their qualifications match the job requirements. Candidates are ranked according to their suitability scores and the system will provide explainable breakdown of the evaluation criteria to improve transparency in candidate selection. Additionally, the system assist HR in recruitment communication by generating customizable email drafts for candidate notifications. The proposed system functions as a

decision support tool that enhance recruitment efficiency while maintaining human oversight in hiring decision.

(272 words)

(ii)	Detailed Proposal of the Project
	(a) Introduction <i>(Project Background and Problem Statements)</i>
	Project Background In order to maintain competitiveness and support innovation, organization increasingly depend on skilled knowledge workers. As a result, recruitment become a crucial strategic process aimed to identifying candidates who possess the appropriate qualifications required for specific roles. The adoption of online recruitment platforms has enabled organizations to reach wider pool of applicants, however it also resulted in a significant increase in the number of job applications received. Human resource departments that hold wide range of responsibilities including employee relations, payroll management, training coordination and recruitment activities may experience difficulties to efficiently screen large volumes of resumes within limited timeframes. Furthermore, modern hiring practices often include collaborative decision-making where multiple stakeholders participate in evaluating candidates. This collaborative process requires structured and transparent methods of assessing candidate suitability. Although many online job platforms provide automated candidate suggestions, these recommendation are often based on generalized matching algorithms and may not accurately capture the specific requirements described in job postings. Consequently, recruiters still need to manually examine numerous resumes to determine candidate suitability. Therefore, this project proposes the development of an intelligent system that able to assist HR recruiters in screening and ranking candidates by analyzing job descriptions and resumes in a structured and explainable manner. Problem Statements Based on the background discussed, the following problems have been identified: 1. Organizations often receive a large number of applications or a single job vacancy, making manual resume screening time-consuming and inefficient. 2. HR personnel handle various organizational responsibilities which limits the time available to manually review and evaluate large numbers of resumes especially when hiring needs are urgent. 3. Automated candidate suggestions provided by recruitment platforms may not accurately match the specific requirements stated in job descriptions. 4. Resume submitted often comes in many formats and structures making it difficult to objectively compare candidate qualifications with job requirements.
	(b) Objectives of the Project
	This project embarks on the following objectives: 1. To design a job description analysis module that extracts key job requirements such as skills, educations and experience.

2. To develop a resume parsing mechanism that converts unstructured resume documents into structured candidate information.
3. To implement a weighted similarity scoring model that evaluates candidate suitability based on job requirements.

(c) Scope of the Project

The following is the proposed project scope:

Target Users

1. Human Resource personnel
2. Hiring managers
3. Job applicants

Platform Scope

1. Developed as a web based application accessible through modern web browsers.
2. Implemented using standard web technologies for both front end and back end development.
3. Functions as a standalone prototype designed for demonstration and academic purposes rather than commercial deployment.

Functional Scope

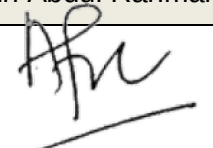
1. Allow hiring managers to upload or input job descriptions to define job requirements
2. Allow candidate to submit applications through a dedicated application link, including personal details and resume uploaded
3. Extract and process information from uploaded resumes such as skills, education and work experience.
4. Compare candidate information with job requirements using a weighted matching mechanism
5. Generate and display ranked candidate recommendation for HR review
6. Provide an explainable breakdown showing how each candidate's score was calculated
7. Allow HR users to generate email draft templates for candidate communication such as rejection or interview invitation.

Data Scope

1. Job description data provided by HR users to define job requirements and evaluation criteria.
2. Candidate application data submitted through the system including personal details and uploaded resumes.
3. Resume documents limited to text-based PDF formats containing information such as skills, education and work experience.
4. Extracted resume information processed into structured data for candidate-job matching and ranking.

	Limitations 1. The system focus only on resume screening and candidate shortlisting stage of recruitment 2. Does not perform any psychological assessments, interview analysis or final hiring decision, which remain the responsibilities of the HR personnel and hiring managers 3. Resume processing is limited to text-based PDF document formats 4. The system evaluates candidates based only on information provided in application forms and resumes
	(d) Expected Outcome/ Proposed Solution 1. A web-based intelligent resume screening system that assists HR recruiters in managing and analyzing job applications. 2. An automated resume processing mechanism that able to converts unstructured resume documents into structured candidate information. 3. A candidate evaluation model that ranks applicants based on their similarity to job requirements using weighted scoring techniques. 4. An explainable ranking interface that improves transparency in recruitment decisions. 5. A centralized application submission platform that able to automatically collect candidate resumes through system-generated application links. 6. A recruitment support features that allows HR personnel to generate customizable email drafts for communicating with candidates.
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<i>State your references (Minimum 10 references)</i>	
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3	Lussier, R. N., & Hendon, J. R. (2025a). <i>Human Resource Management</i>. Sage Publications.
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5	Okatta, C. G., Ajayi, F. A., & Olawale, O. (2024). <i>NAVIGATING THE FUTURE: INTEGRATING AI AND MACHINE LEARNING IN HR PRACTICES FOR A DIGITAL WORKFORCE</i>. Computer Science & IT Research Journal, 5(4), 1008–1030. https://doi.org/10.51594/csitrj.v5i4.1085
6	Soleimani, M., Intezari, A., Arrowsmith, J., Pauleen, D. J., & Taskin, N. (2025). <i>Reducing AI bias in recruitment and selection: an integrative grounded approach</i>. The International Journal of Human Resource Management, 36(14), 1–36. https://doi.org/10.1080/09585192.2025.2480617

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8	Vanderpal, G., & Brazie, R. (2022a). The Impact of Underlying Stress and Trauma on HRM Recruitment and Selection Bias in Employee Interviews. <i>Journal of Leadership, Accountability and Ethics</i> , 19(2). https://doi.org/10.33423/jlae.v19i2.5111
9	Vivek, R. (2022a). Is Blind Recruitment an Effective Recruitment Method? <i>International Journal of Applied Research in Business and Management</i> , 3(3), 56–72. https://doi.org/10.51137/ijarbm.2022.3.3.4
10	Wörsdörfer, M. (2023). Mitigating the adverse effects of AI with the European Union's artificial intelligence act: Hype or hope? <i>Global Business and Organizational Excellence</i> , 43(3). https://doi.org/10.1002/joe.22238

E DECLARATION BY STUDENT AKUAN PELAJAR	
(i)	Date: 02 April 2026 Student's Signature: 
E RECOMMENDED BY SUPERVISOR PERAKUAN OLEH PENYELIA	
	Recommended ☒ Not Recommended ☐
	Comments: Student has discussed the proposal and done some amendments based on the discussion.
	Supervisor's Name: Ts Ahmad Fadzli Nizam bin Abdul Rahman
	Signature & Stamp:  AHMAD FADZLI NIZAM BIN ABDUL RAHMAN Pensyarah Kanan Fakulti Teknologi Maklumat dan Komunikasi Universiti Teknikal Malaysia Melaka (UTeM)
	Date: 3 April 2026